

Faculty of Artificial Intelligence

UTM UNIVERSITI TEKNOLOGI MALAYSIA

TOPIC 5: UNSUPERVISED LEARNING

SAIA 2113 – Machine Learning
Session 1 2025/2026

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Innovating Solutions

Topic 5: Unsupervised Learning

- Introduction - What is unsupervised learning?
- The language of unsupervised learning
- Unsupervised learning in practice

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Introduction - What is unsupervised learning?

Unsupervised learning utilizes dataset that does not contain labels.

Unsupervised learning aims to discover interesting and useful patterns in data:

- Clusters of related data points. For example, we might want to discover groups of similar customers from the logs of an e-commerce website.
- Outliers, i.e., particularly unusual or interesting data points. For example, suspicious financial transactions.
- Denoised signals. Recovering an image corrupted with white noise.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

Unsupervised Learning: An Example

- Unsupervised learning can be described via the following recipe:



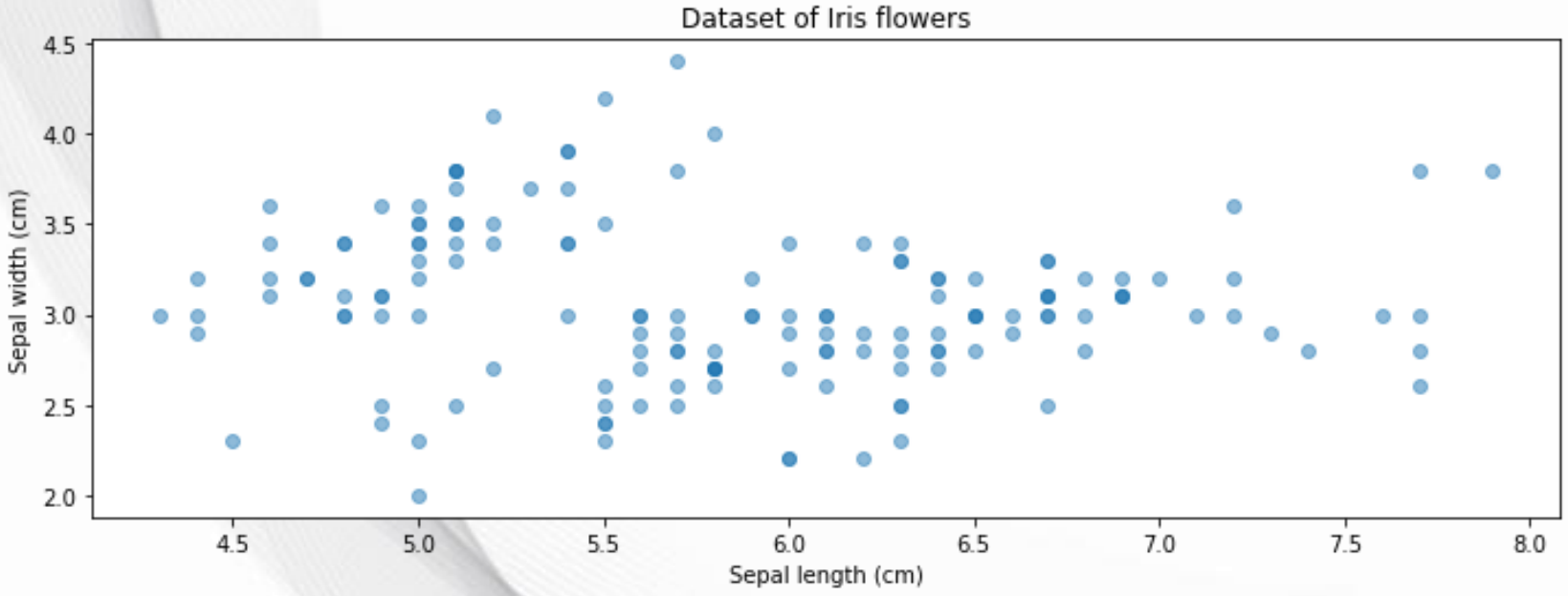
- The unsupervised model is not a predictive model.
- Rather, it describes interesting structure in the data.

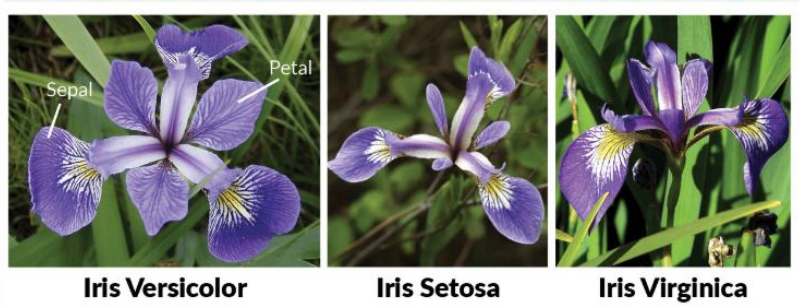
Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

An Unsupervised Learning Dataset

- The Iris flower dataset will be used. All the labels from this dataset will be discarded to make things truly unsupervised.

- The petal length and width is discarded. Only the sepal length and width are used as the attributes for each datapoint. The class label are also discarded.



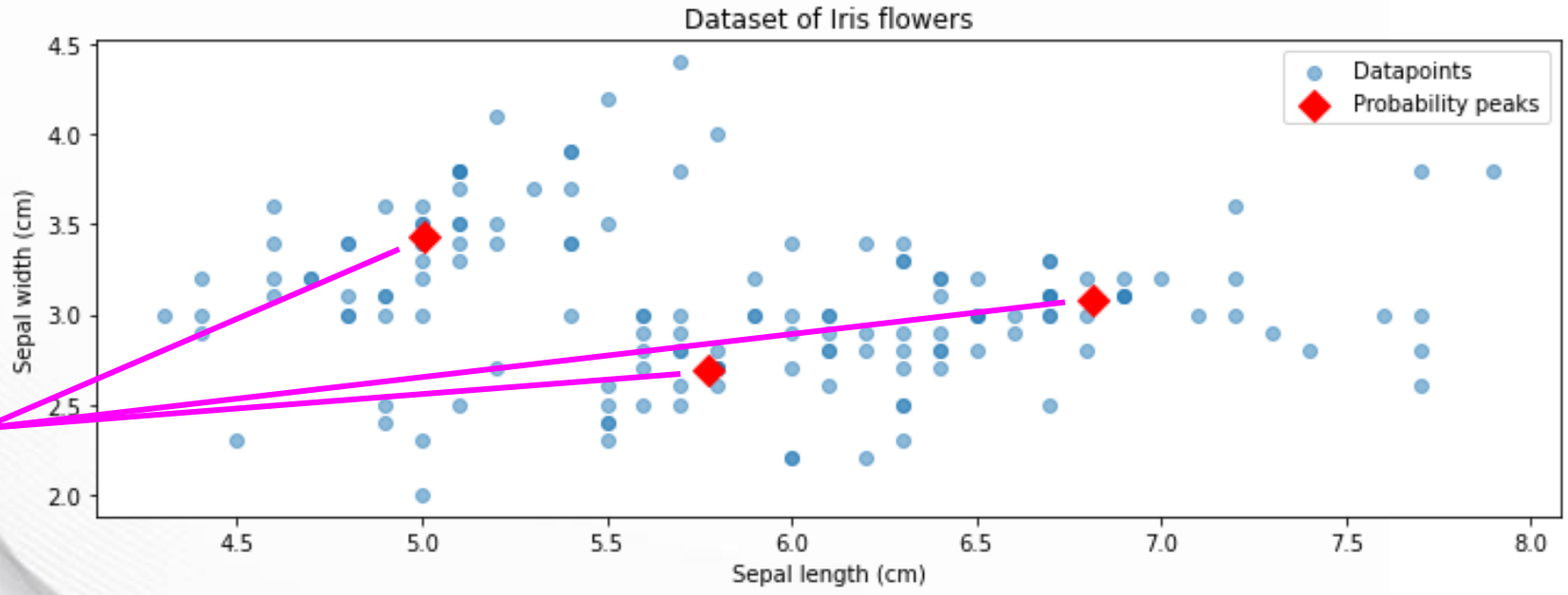


Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

An Unsupervised Learning Algorithm

- A popular unsupervised learning algorithm, **K-Means** will be used with the Iris flower dataset.
- The algorithm seeks to find K clusters in the data.
- Each cluster is characterized by its centroid (its mean).

The red diamonds indicate the locations of the three cluster centroids.



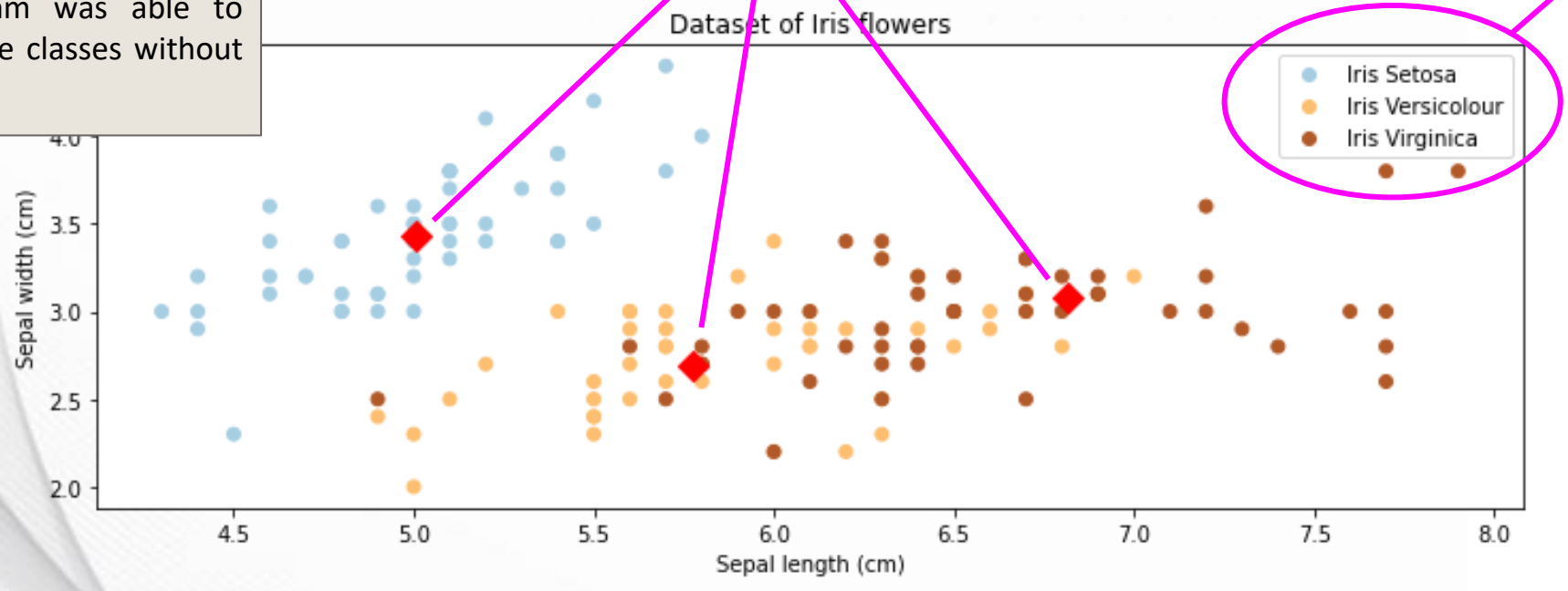
Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

An Unsupervised Learning Algorithm

The centroids approximately correspond to the three classes of flowers present in the dataset.

The class labels are leveraged to understand the centroids provided by the K-means algorithm.

K-means algorithm was able to discover the three classes without using any labels.



Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

Applications of Unsupervised Learning

Red words are related to biology.

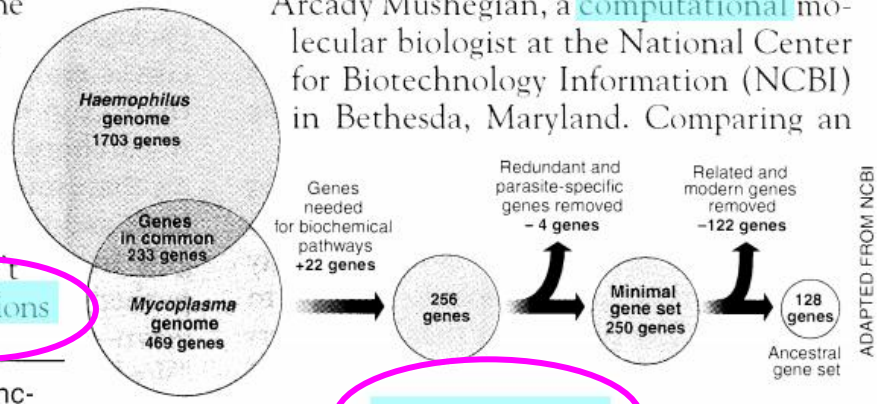
Yellow words pertain to genetics.

Blue words are related mostly to computers.

Seeking Life's Bare (Genetic) Necessities

COLD SPRING HARBOR, NEW YORK—How many genes does an organism need to survive? Last week at the genome meeting here,* two genome researchers with radically different approaches presented complementary views of the basic genes needed for life. One research team, using computer analyses to compare known genomes, concluded that today's organisms can be sustained with just 250 genes, and that the earliest life forms required a mere 128 genes. The other researcher mapped genes in a simple parasite and estimated that for this organism, 800 genes are plenty to do the job—but that anything short of 100 wouldn't be enough.

Although the numbers don't match precisely, those predictions "are not all that far apart," especially in comparison to the 75,000 genes in the human genome, notes Siv Andersson of Uppsala University in Sweden, who arrived at the 800 number. But coming up with a consensus answer may be more than just a genetic numbers game, particularly as more and more genomes are completely mapped and sequenced. "It may be a way of organizing any newly sequenced genome," explains Arcady Mushegian, a computational molecular biologist at the National Center for Biotechnology Information (NCBI) in Bethesda, Maryland. Comparing an



* Genome Mapping and Sequencing, Cold Spring Harbor, New York, May 8 to 12.

Stripping down. Computer analysis yields an estimate of the minimum modern and ancient genomes.

SCIENCE • VOL. 272 • 24 MAY 1996

Image source: Probabilistic Topic Models by David M. Blei

Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

Applications of Unsupervised Learning

Unsupervised topic models automatically discover clusters of related words.

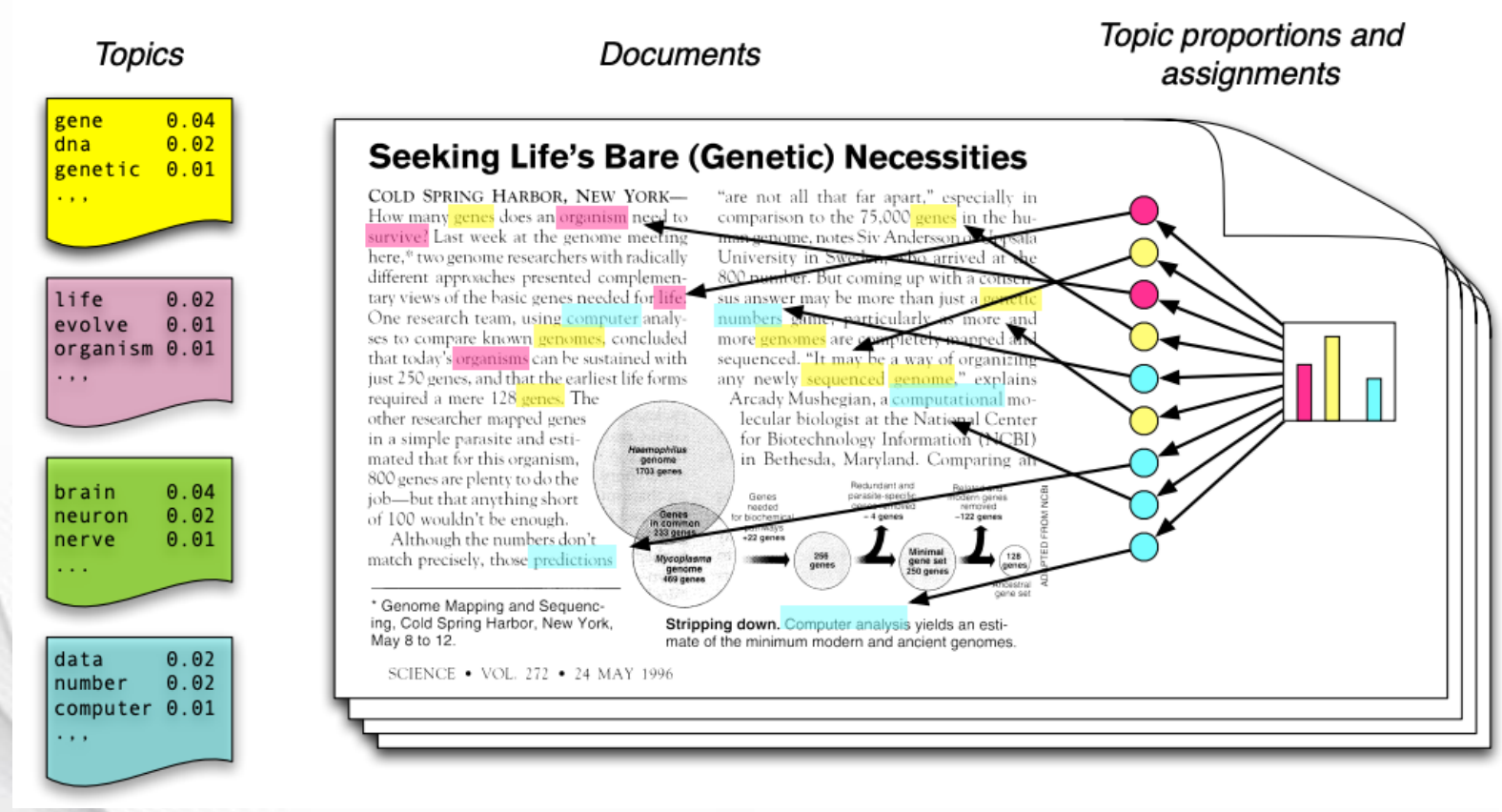


Image source: Probabilistic Topic Models by David M. Blei

Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

Applications of Unsupervised Learning

Other use cases for unsupervised learning include:

1. **Product defect detection in manufacturing for quality control.**

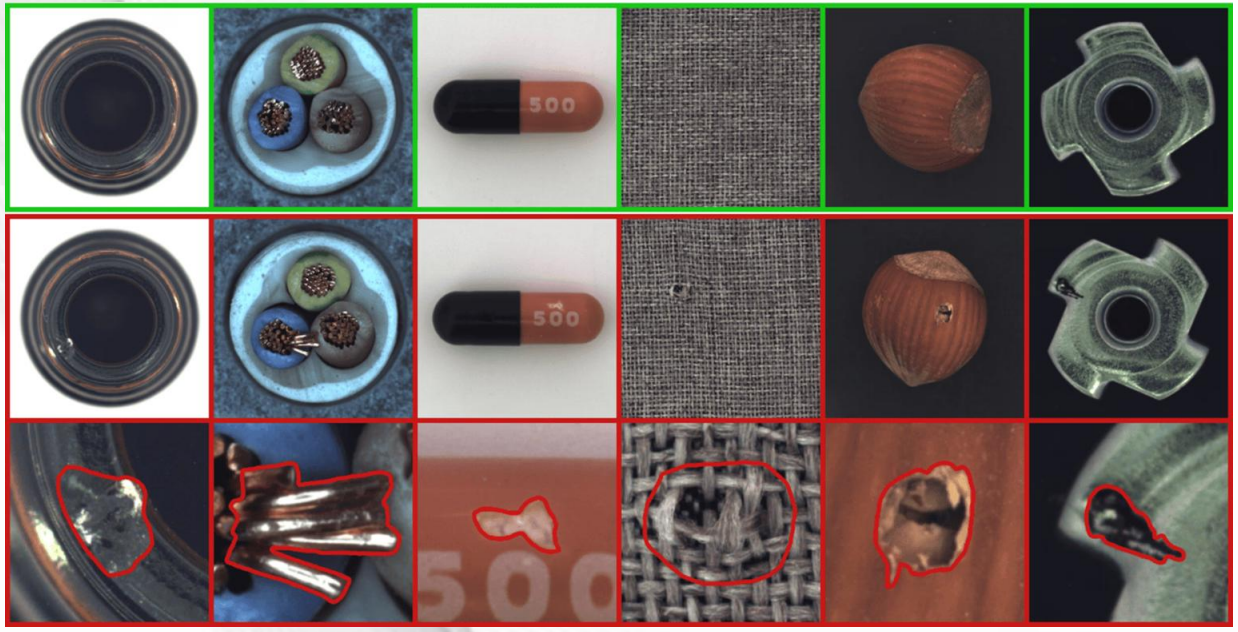


Image Source: https://superlinear.eu/insights/articles/efficient-product-defect-detection-in-manufacturing-unsupervised-anomaly-detection-for-quality-control

2. **Signal denoising.**

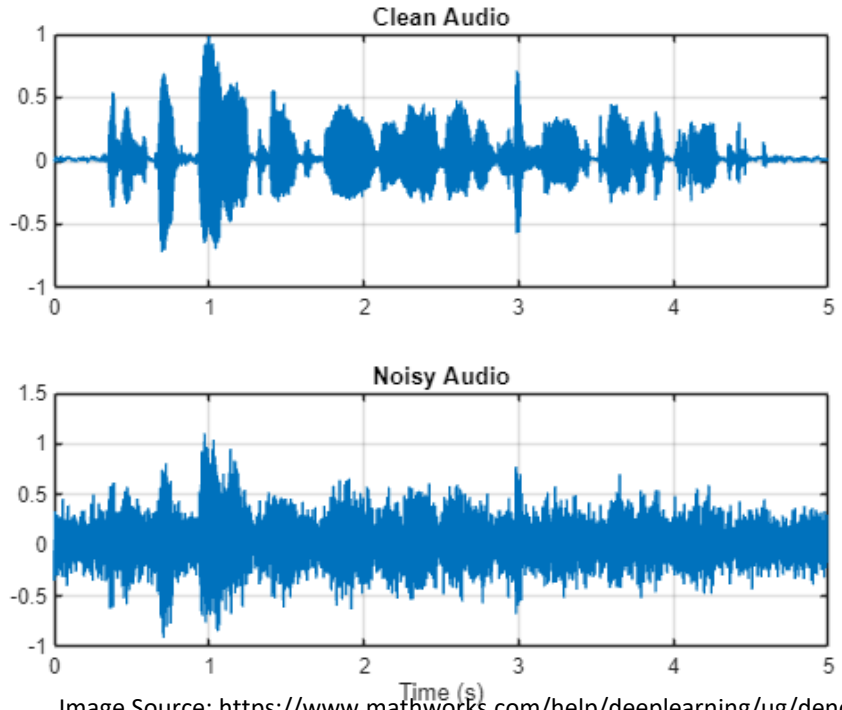


Image Source: https://www.mathworks.com/help/deeplearning/ug/denoise-speech-using-deep-learning-networks.html

Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

Topic 5: Unsupervised Learning

- Introduction - What is unsupervised learning?
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The language of unsupervised learning

An unsupervised machine learning problem can be described by using the following “recipe”:

Dataset + Learning Algorithm → Unsupervised Model

Attributes, Features Model Class + Objective + Optimizer

More formally, an unsupervised dataset of size *n*, can be defined as:

$$\mathcal{D} = \left\{ x^{(i)} \middle| \, i = 1, 2, \ldots, n \right\}$$

Each $x^{(i)} \in \mathbb{R}^d$ denotes an input, a vector of *d* attributes or features.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

The language of unsupervised learning

Unsupervised learning algorithm consisted of three components:

- A model class: the set of possible unsupervised models we consider.
- An objective function, which defines how good a model is.
- An optimizer, which finds the best predictive model in the model class according to the objective function.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

Model & Model Class

We'll say that a model is a function

$$f_\theta : \mathcal{X} \rightarrow \mathcal{S}$$

that maps inputs $x \in \mathcal{X}$ to some kind of structure $s \in \mathcal{S}$. Models may have parameters $\theta \in \Theta$ living in a set Θ.

Formally, the model class is a set

$$\mathcal{M} = \{ f_\theta \mid f_\theta : \mathcal{X} \rightarrow \mathcal{S}; \theta \in \Theta \}$$

of possible models (models with different parameters) that map input features to structural elements.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

Learning Objective

An *objective function* (also called a *loss function*) is defined as:

$$J(\theta) : \Theta \rightarrow [0, \infty),$$

which describes the extent to which f_θ "fits" the data

$$\mathcal{D} = \left\{ x^{(i)} \middle| \, i = 1, 2, \ldots, n \right\}$$

Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

Optimizer

An optimizer finds a model $f_\theta \in \mathcal{M}$ with the smallest value of the objective J.

$$\min_{\theta \in \Theta} J(\theta)$$

Intuitively, this is the function that bests “fits” the data on the training dataset.

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An Example: K-Means

The K-means algorithm is derived to better understand how these three components (model class, objective function, optimizer) define an unsupervised learning algorithm.

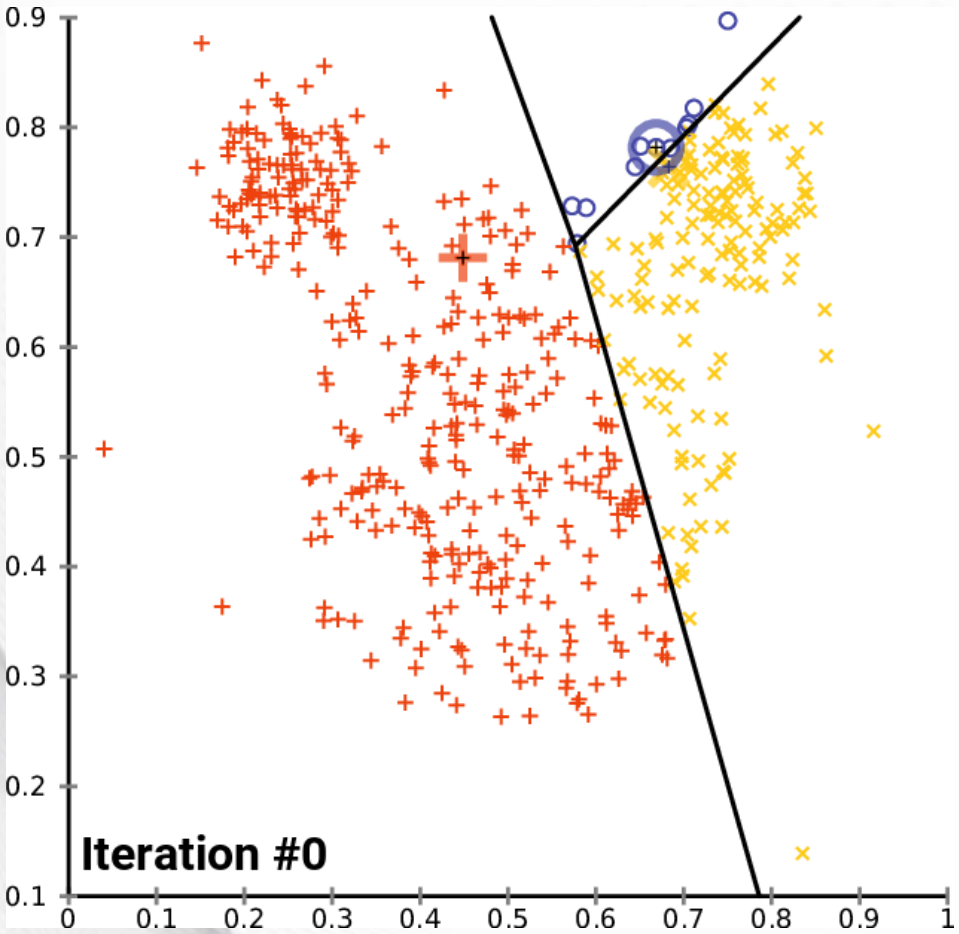
At a high level, K-means performs the following steps. Starting from random centroids, we repeat until convergence.

1. Update each cluster: assign each point to its closest centroid.

2. Set each centroid to be the center of the its cluster.

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An Example: K-Means



The big red "+", yellow "x", and blue "o" symbols represent clusters.

At each iteration, they get recomputed to the center of the red, yellow, and blue points respectively.

After they are recomputed, the data points are re-colored to match the color of their closest centroid.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

The K-Means Model

- The parameters θ of the model are K centroids $c_1, c_2, \ldots c_K \in \mathcal{X}$. The class of x is going to be k if c_k is the closest centroid to x. (Assume the distance metric in use is Euclidean distance).

- We can think of the model returned by K-Means as a function

$$f_\theta(x) = \arg\min_k \|x - c_k\|$$

that assigns each input x to a cluster $s \in \mathcal{S} = \{1, 2, \ldots, K\}$.

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The K-Means Objective

- How do we determine whether f_θ is a good clustering of the dataset *D*?
- We seek centroids c_k such that the distance between the points and their closest centroid is minimized:

$$J(\theta) = \sum_{i=1}^{n} \left\| x^{(i)} - \text{centroid}\left(f_\theta\left(x^{(i)}\right)\right) \right\|,$$

where centroid $(k) = c_k$ denotes the centroid for cluster k.

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The K-Means Optimizer

The K-Means algorithm can be optimized in a two-step process, stating with an initial random cluster assignment $f_\theta(x)$.

Starting with random centroids c_k, repeat until convergence:

1. Update such that $f(x)$ such that $f_\theta(x) = \arg\min_k \|x - c_k\|$ is the cluster of the closest centroid to $x^{(i)}$.

2. Set each c_k to be the center of its cluster $\{x^{(i)} | f(x^{(i)}) = k\}$.

This process is guaranteed to converge after a finite number of iterations. The intuition is that the objective function decreases at each step, hence at some point it must reach an optimum.

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Algorithm: K-Means

A model card for K-Means can be provided in terms of its standard components:

- **Type:** Unsupervised learning (clustering).
- **Model family:** k centroids.
- **Objective function:** Sum of distances (of your choice) to the closest centroids.
- **Optimizer:** Iterative optimization procedure.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

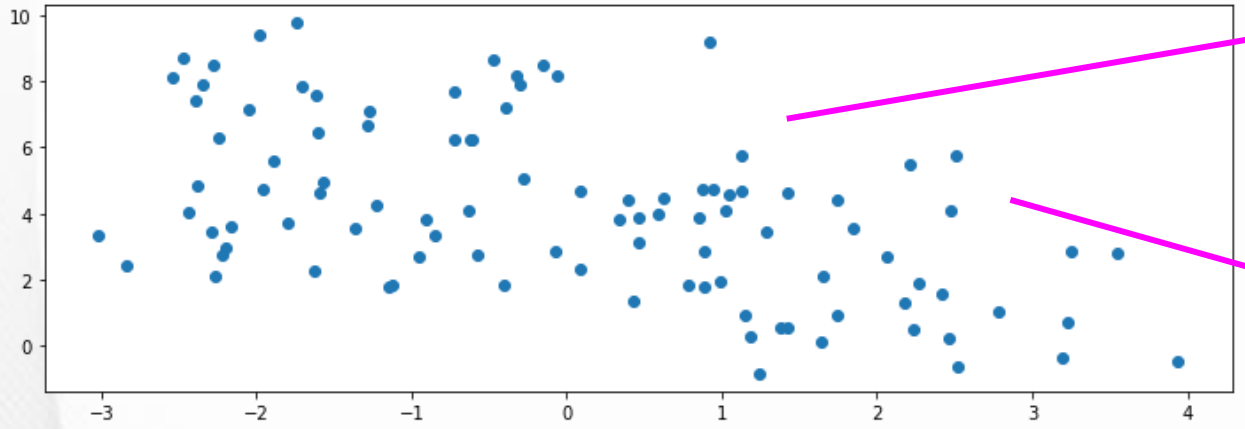
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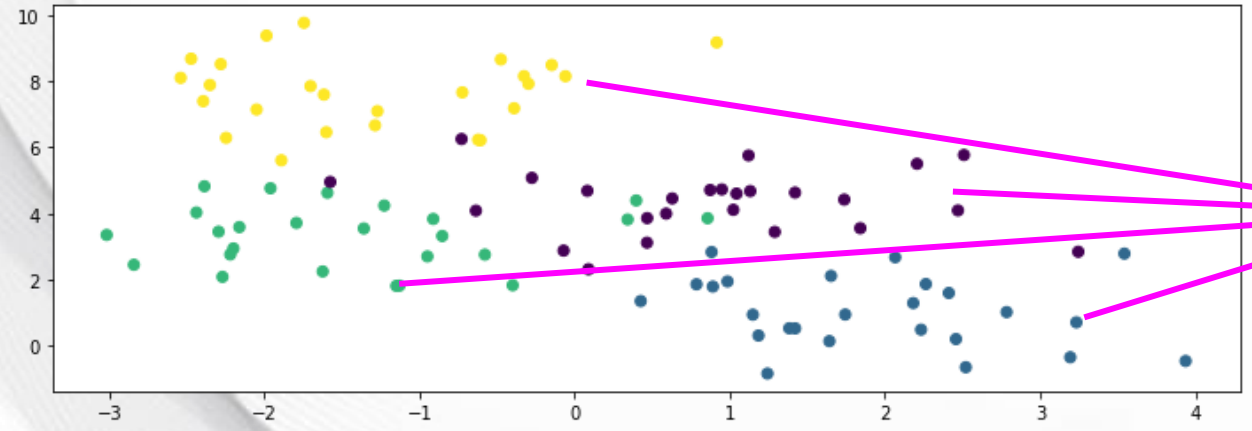
Unsupervised learning in practice

An Unsupervised Learning Dataset:



The 100 data points are generated by a mixture of four Gaussian distributions.

A sample from this mixture distribution should form roughly four clusters.



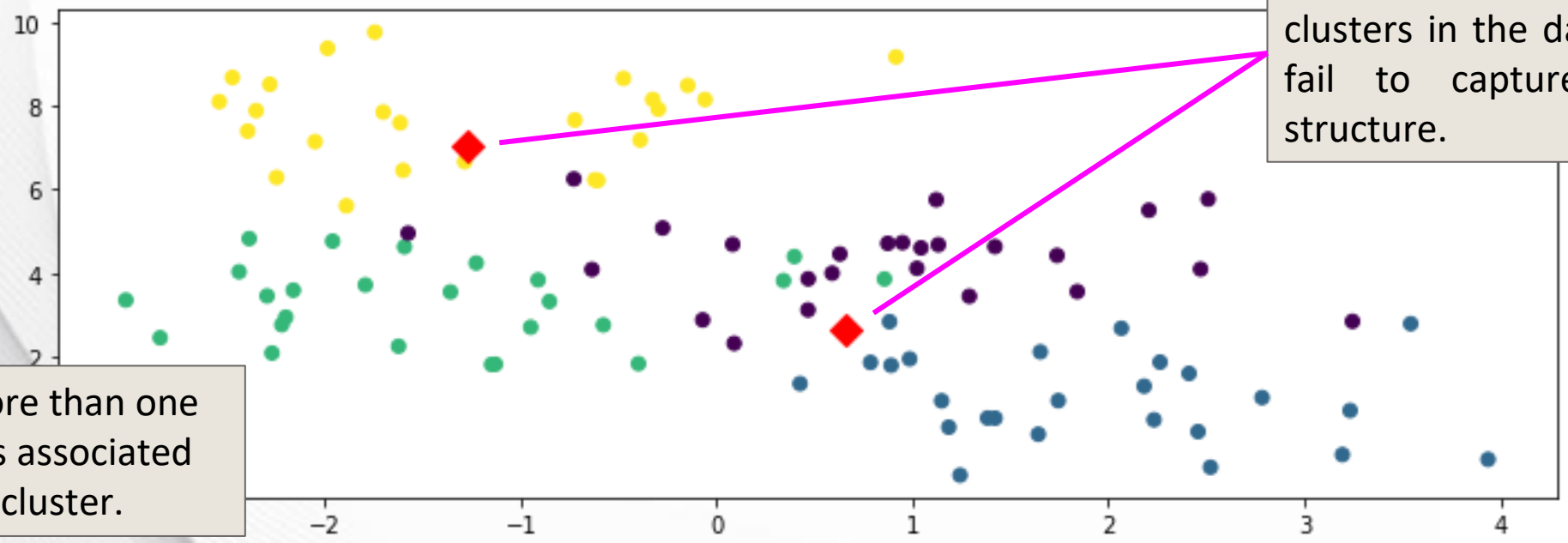
These four clusters are visualized with different colors based on the true labels.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

Underfitting in Unsupervised Learning

- In supervised learning, underfitting happens when we are not able to fully learn the signal present in the data.

- In the context of K-Means, this means our K is lower than the actual number of clusters in the data.



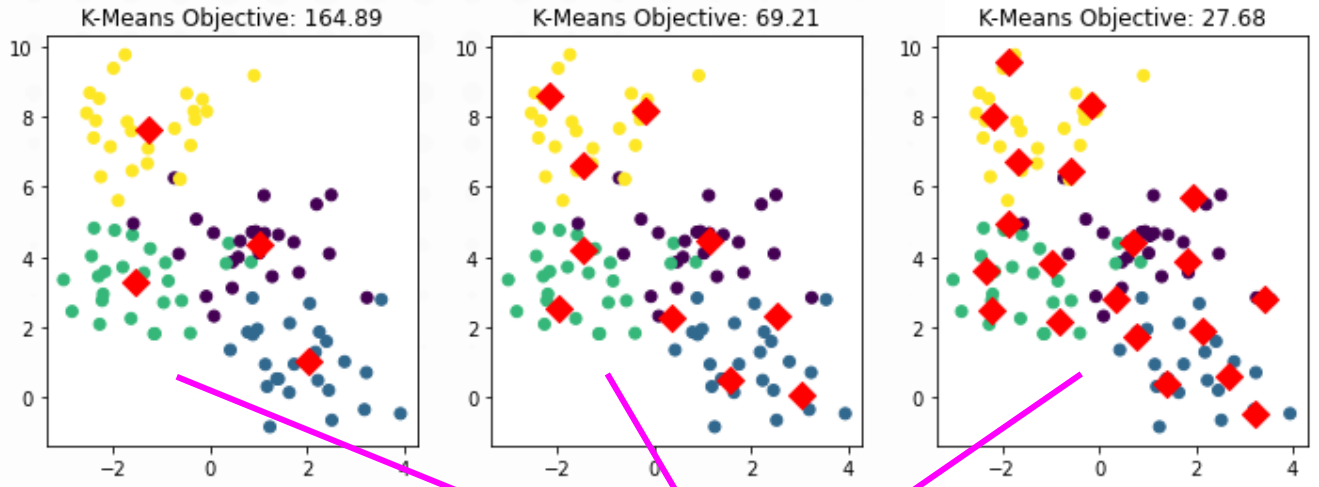
The centroids find two clusters in the data, but they fail to capture the true structure.

There are more than one true cluster is associated with learned cluster.

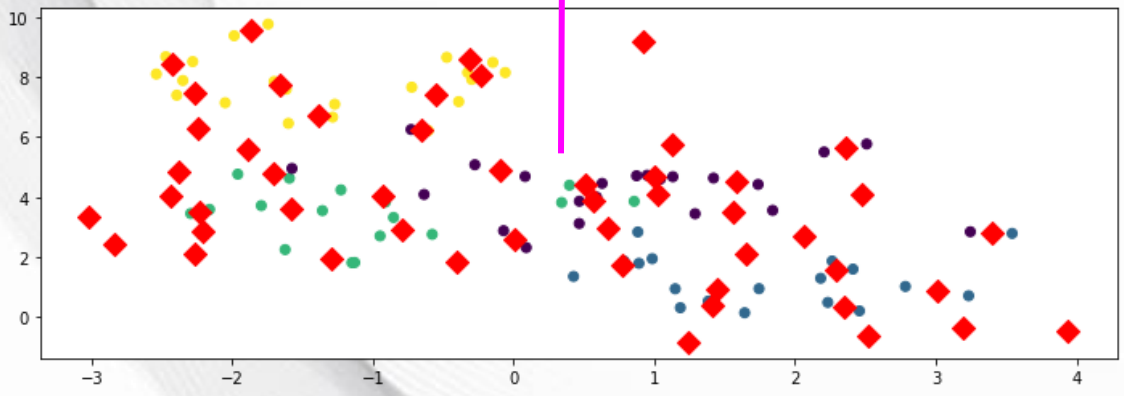
Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

Overfitting in Unsupervised Learning

- Overfitting happens when we fit the noise, but not the signal.
- This means fitting small, local noise clusters rather than the true global clusters.



The K is set to 50.



The K is set to 4, 10, and 20 from left to right respectively.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

Generalization in Unsupervised Learning

In generalization, it is assumed that the dataset is sampled from a probability distribution $\mathbb{P}$, which we will call the data distribution.

$$x \sim \mathbb{P}.$$

It is assumed that the dataset $\mathcal{D} = \left\{x^{(i)} \middle| i = 1, 2, \ldots, n\right\}$ consists of independent and identically distributed (IID) samples from $\mathbb{P}$.

What independent and identically distributed (IID) means is:

- Each training example is from the same distribution.
- This distribution doesn't depend on previous training examples.

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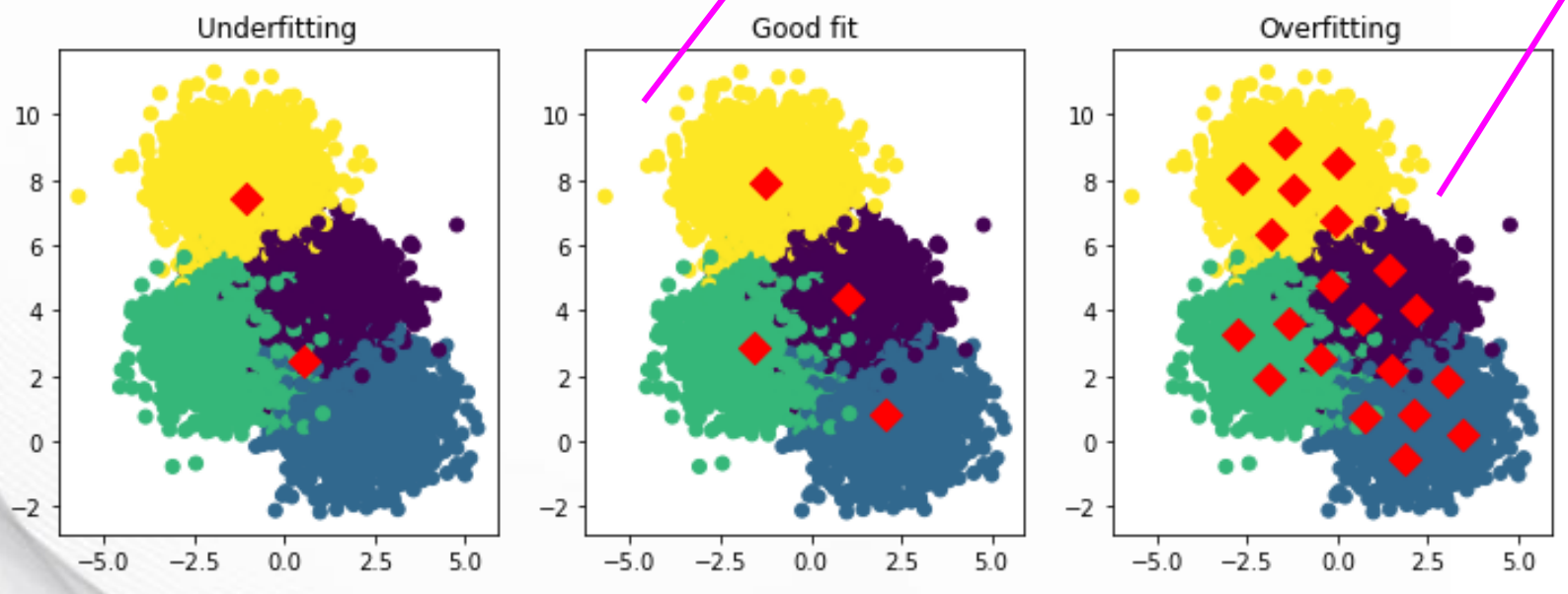
Generalization in Unsupervised Learning

We can think of the data distribution as being the sum of two distinct components, $\mathbb{P} = F + E$

1. A signal component F (hidden clusters, speech, low-dimensional data space, etc.)
2. A random noise component *E.*

ML model generalizes if it fits the true signal F.

ML model overfits if it learns the noise *E*.



Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

Detecting Overfitting and Underfitting

In real-life scenarios when data is not in two-dimensional, visualizing the data and the cluster centres might not help us to detect overfitting or underfitting. We also may not have the access to the true cluster labels.

Generally, in unsupervised learning, overfitting and underfitting are more difficult to quantify than in supervised learning due to several reasons:

- Performance may depend on our intuition and require human evaluation.
- If we know the true labels, we can measure the accuracy of the clustering. But we do not have the labels for unsupervised learning.

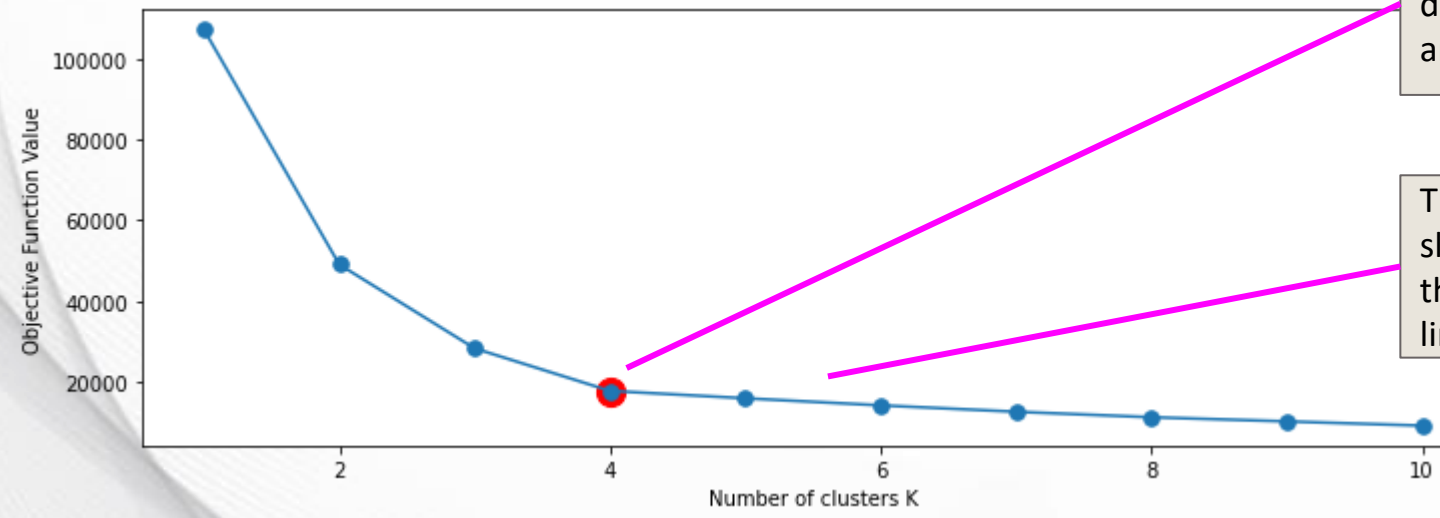
If our model is probabilistic, in order to detect overfitting without labels is to compare the log-likelihood between the training set and a holdout set.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

The Elbow Method

The elbow method is a useful heuristic that can be used to tune hyperparameters in unsupervised learning. E.g., choosing K for K-means. The elbow method works as follows:

- Plot the objective function as a function of the hyper-parameter K.
- The “elbow” of the curve happens when its rate of decrease substantially slows down.
- The “elbow” is a good guess for the hyperparameter.



When K = 4, the objective value does not improve too much anymore.

The decrease in objective values slows down after K = 4. After that, the curve becomes just a line.

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Reducing Overfitting

There are multiple ways to control overfitting including:

1. Reduce model complexity (e.g., reduce K in K-Means)
2. Penalize complexity in the objective (e.g., penalize large K)
3. Use a probabilistic model and regularize it.

Source: https://kuleshov-group.github.io/aml-book/contents/lecture8-unsupervised-learning.html#the-language-of-unsupervised-learning

Generalization in Unsupervised Learning: Summary

- In supervised learning, it is easier to quantify via accuracy.
- In unsupervised learning, we may not be able to easily detect overfitting, but it still happens.

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THANK YOU

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